PROOF OF CONCEPT DOCUMENT

QR-BASED CONSTRUCTION SITE ENTRY & EXIT VERIFICATION SYSTEM

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1. POC PURPOSE

The purpose of this Proof of Concept (POC) is to demonstrate a QR-based system for verifying and recording the entry and exit of employees and visitors at construction sites.

The system will allow employees to verify their identities through a mobile-friendly website accessed by scanning a QR code. Their attendance will be recorded automatically and displayed on an administrative web dashboard.

For individuals visiting the site for the first time, the system will provide a visitor registration option to collect basic information.

The initial POC will focus on employee verification, visitor registration, attendance recording, and administrative monitoring.

2. SYSTEM OVERVIEW

The system will consist of three main components:

- **QR Verification Website:** Used by employees and visitors to verify their identities or register their details.
- **Administrative Web Dashboard:** Used by owners and administrators to manage employees and monitor site attendance.
- **Backend and Database:** Responsible for identity verification, attendance processing, and data storage.

The system will support the following user roles:

- Super Administrator
- Site Owner / Administrator
- Employee
- New Visitor

The application will be accessible through a mobile browser, so employees and visitors will not need to install a separate mobile application.

3. USER ROLES

3.1 Super Administrator

The Super Administrator will manage the overall platform, including administrator accounts, construction sites, and platform-level settings.

3.2 Site Owner / Administrator

The owner or administrator will be able to:

- Register and manage employees.
- Assign employee IDs or access codes.
- View attendance records.
- Monitor current site attendance.
- View visitor registrations.
- Search and filter attendance history.

3.3 Employee

Employees will be able to scan the site QR code, enter their assigned ID or access code, verify their identity, and record their entry or exit.

3.4 New Visitor

New visitors will be able to scan the QR code, select the visitor registration option, enter their basic information, and submit their details. Their registration and approval status will be available to the administrator.

4. PHASE 1 — QR VERIFICATION & ATTENDANCE MANAGEMENT

Phase 1 will focus on the essential entry and exit verification workflow.

4.1 QR Code Access

A QR code will be displayed at the construction site entrance. Scanning the code will open a website containing two options:

- **Registered Employee:** Enter employee ID or access code.
- **New Visitor:** Register basic personal information.

The QR code will identify the relevant construction site.

4.2 Employee Verification

Registered employees will enter their assigned ID or access code. The system will validate the information against the employee records.

If verification is successful, the system will determine whether to record an entry or exit based on the employee's current attendance status.

The system will display a confirmation message after successfully recording the event.

Example:

Entry Verified Successfully

Employee: John Silva

Entry Time: 08:05 AM

Your entry has been recorded successfully.

Invalid credentials or inconsistent attendance actions should generate appropriate error messages.

4.3 Visitor Registration

Visitors without an employee ID can select the New Visitor option and submit basic information, including:

- Full name
- Contact number
- Organization or company, if applicable
- Purpose of visit

The system will create a visitor record and display the registration status.

Where required, the administrator can review and approve the visitor before site access is authorized. Visitor departure should also be recorded using an appropriate visitor reference or verification method.

5. ENTRY & EXIT WORKFLOW

Employee Workflow

Scan QR Code

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Open Verification Website

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Enter Employee ID / Access Code

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Verify Identity

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Determine Entry or Exit

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Record Attendance

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Display Confirmation

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Update Admin Dashboard

New Visitor Workflow

Scan QR Code

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Select New Visitor

↓

Enter Basic Information

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Submit Registration

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Create Visitor Record

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Apply Approval Rules, If Required

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Update Admin Dashboard

6. ADMINISTRATIVE WEB DASHBOARD

The dashboard will provide a centralized view of construction site attendance and visitor activity.

6.1 Dashboard Overview

The dashboard may display:

- Total registered employees
- Employees currently recorded as present
- Total entries and exits for the day
- Visitor registrations
- Employees with incomplete attendance sessions

6.2 Employee Management

Administrators will be able to register employees, assign IDs or access codes, update employee details, activate or deactivate accounts, and view individual attendance history.

6.3 Attendance Monitoring

The attendance section will display employee information, site details, entry time, exit time, and attendance status.

Administrators should be able to search and filter records by employee, date, site, and status where applicable.

6.4 Visitor Management

Administrators will be able to view visitor details, purpose of visit, registration time, approval status, and departure information where recorded.

7. SYSTEM ARCHITECTURE

The system will follow a centralized architecture.

QR Code

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Mobile-Friendly Verification Website

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Backend / API

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Centralized Database

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Administrative Web Dashboard

The backend will handle identity verification, attendance processing, visitor registration, and authorization. The dashboard will display information retrieved from actual system records.

8. CORE DATA REQUIREMENTS

The system will maintain the following main data entities:

- **Construction Site:** Site ID, name, location, and status.
- **Employee:** Employee ID, name, contact details, assigned site, access credentials, and account status.
- **Visitor:** Visitor reference, name, contact details, organization, purpose of visit, and approval status.
- **Attendance Record:** Employee or visitor reference, site, entry time, exit time, and attendance status.
- **Administrator:** Account details, role, permissions, and assigned site.

9. SECURITY REQUIREMENTS

The system should implement:

- Secure administrator authentication.
- Role-based access control.
- Backend validation of employee IDs and access codes.
- Protection of employee and visitor information.
- Prevention of duplicate or inconsistent attendance records.
- Secure communication between the website and backend.
- Appropriate logging of important administrative actions.

The QR code should provide access to the verification page but should not be treated as proof of identity or physical presence at the site.

10. POC SCOPE

The initial POC will include:

- QR code access for a construction site.
- Mobile-friendly verification website.
- Employee ID or access-code verification.
- Entry and exit attendance recording.
- New visitor registration.
- Basic visitor approval handling.
- Administrator login.
- Employee management.
- Attendance dashboard and history.
- Basic search and filtering.
- Verification confirmation and error messages.

The POC will demonstrate the complete workflow rather than a full enterprise workforce management solution.

11. FUTURE ENHANCEMENTS

The platform may later be extended with:

- Multiple construction sites.
- Rotating QR codes.
- GPS-based location verification.
- Employee photo verification.
- Contractor and subcontractor management.
- Visitor pre-registration.
- Advanced attendance reports.
- Payroll and shift management.
- Safety induction tracking.
- Emergency evacuation lists.

12. EXPECTED POC OUTCOME

The POC should demonstrate that employees can verify their identities and record entry and exit times using a QR code, while new visitors can register their information through the same website.

Owners and administrators should be able to monitor attendance, view visitor registrations, and review historical records through the web dashboard.

The resulting system will provide a foundation for a scalable construction site attendance and access monitoring platform.

13. CONCLUSION

The QR-Based Construction Site Entry & Exit Verification System provides a simple digital solution for monitoring workforce attendance and visitor activity.

By combining QR-based verification, employee identification, visitor registration, and a centralized administrative dashboard, the system aims to reduce manual attendance recording and improve visibility into construction site activity.

The initial POC will focus on the core verification and attendance workflows, with additional security and workforce management features available for future development.